

KENDRIYA VIDYALAYA SANGATHAN
CHENNAI REGION
FIRST PRE-BOARD EXAMINATION 2024-25
SCIENCE (086)

Max. Marks: 80

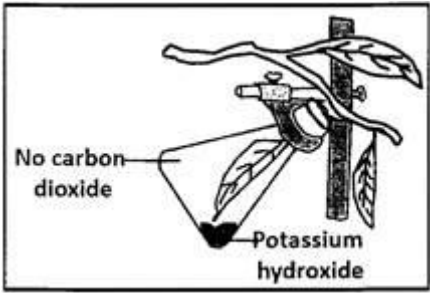
Class: X

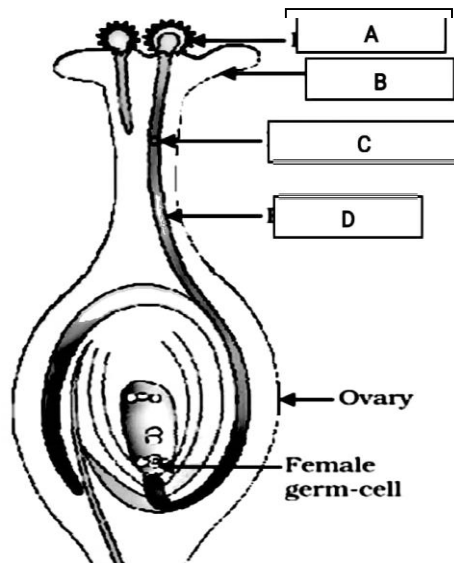
Time Allowed: 3 hours

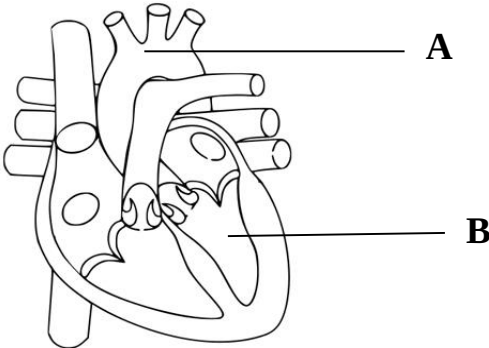
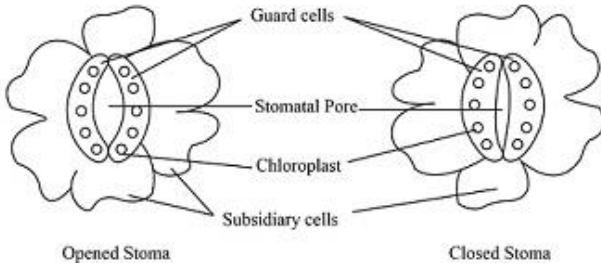
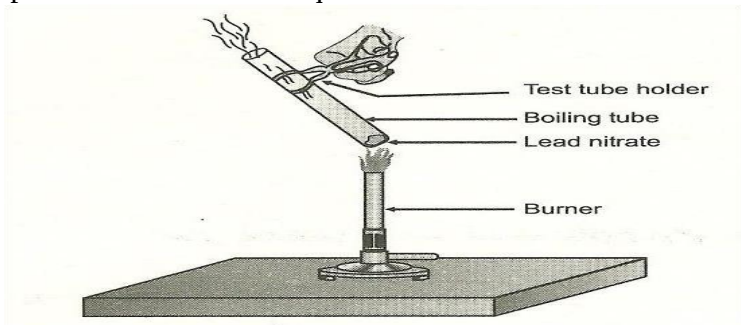
General Instructions:

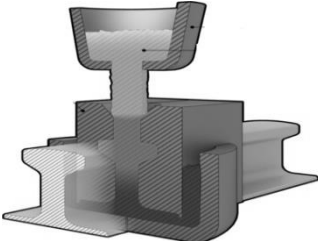
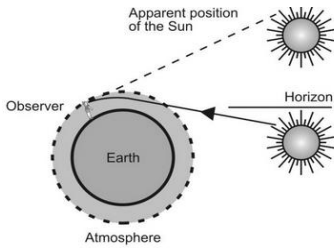
1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

Section- A		
Questions 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.		
1	Out of the following statements, which is correct for the given chemical reaction? $\text{ZnO} + \text{CO} \longrightarrow \text{Zn} + \text{CO}_2$ a) ZnO is being oxidized b) CO is being reduced c) CO ₂ is being oxidized d) ZnO is being reduced	1
2	Baking powder is a mixture of: a) Sodium carbonate and Acetic acid b) Sodium carbonate and Tartaric acid c) Sodium hydrogen carbonate and Tartaric acid d) Sodium hydrogen carbonate and Acetic acid	1
3	From the following reactions which one is considered to be endothermic reaction: a) Decomposition of vegetable matter into compost. b) Decomposition of calcium carbonate to form quick lime and carbon dioxide. c) Burning of a candle d) Process of respiration.	1
4	Identify the following unsaturated hydrocarbons given below? $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$ (i) $\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}-\text{C}=\text{C}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$ (ii) $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ (iii) $\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}_2 \\ \\ \text{Cl} \end{array}$ (iv)	1
a) (i) and (ii) b) (i) and (iv) c) (iii) and (iv) d) (ii) and (iii)		

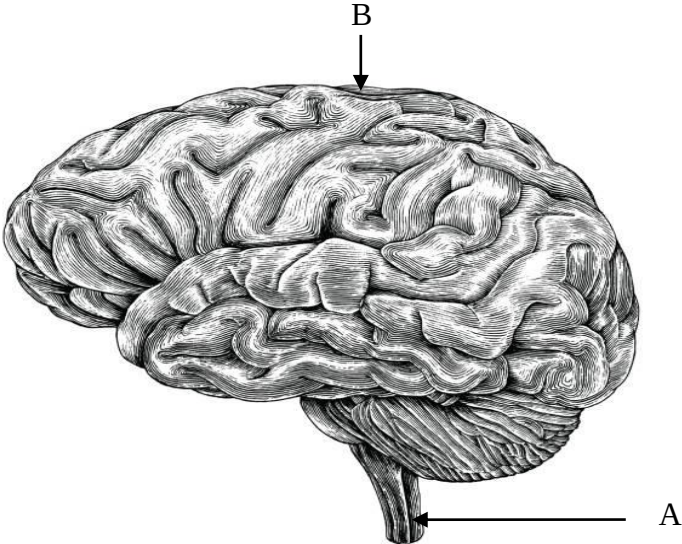
5	Identify the aim of given experiment?  a) To prove that sun light is essential for photosynthesis b) To prove that carbon dioxide is essential for photosynthesis c) To prove that water is essential for photosynthesis d) To prove that chlorophyll is essential for photosynthesis.	1
6	What are the final products of proteins, carbohydrates and fats respectively in human digestion a) (i) Amino acids, (ii) glucose and (iii) fatty acids b) (i) Amino acids, (ii) glucose, (iii) fatty acids and glycerol c) (i) Glucose, (ii) fatty acids and glycerol, (iii) amino acids d) (i) Sugar, (ii) amino acids, (iii) fatty acids and glycerol	1
7	An aqueous solution 'A' turns phenolphthalein solution pink. On addition of an aqueous solution 'B' to 'A', the pink colour disappears. Which of the following statement is true for solution 'A' and 'B'? a) 'A' is strongly basic and 'B' is a weak base. b) 'A' is strongly acidic and 'B' is a weak acid. c) 'A' has pH greater than 7 and 'B' has pH less than 7. d) 'A' has pH less than 7 and 'B' has pH greater than 7.	1
8	Generally, non-metals are not lustrous. Which of the following non-metal is lustrous? a) Sulphur b) Bromine c) Phosphorous d) Iodine	1
9	An object is positioned 30 cm away from a convex mirror with a focal length of 15 cm. At what distance the image is formed from the pole: a) +20cm from the pole b) -10cm from the pole c) +10cm from the pole d) + 60cm from the pole	1
10	Name the plant hormone that acts as a growth inhibitor and is also responsible for causing leaf wilting. a) Absciscic acid b) Cytokinin c) Auxin d) Gibberellin	1
11	Which statement shows interaction of an abiotic component with a biotic component in an ecosystem? a) A mouse fighting with another mouse for food. b) An earthworm making a burrow in the soil. c) A grasshopper feeding on a leaf. d) Rainwater running down into the lake.	1
12	Galvanization is a method of protecting Iron from rusting by coating with a thin layer of: a) Gallium b) Aluminium c) Zinc d) Silver	1
13	What type of energy is transferred from one trophic level to the next in a food chain? a) Heat energy b) Light energy c) Mechanical energy d) Chemical energy	1

14	Identify the type of plant movements from the pictures given below, i)  ii)  a) i) Thigmotropism ii) Hydrotrophism b) i) Hydrotrophism ii) Chemotropism c) i) Hydrotrophism ii) Phototropism d) i) Phototropism ii) Thigmotropism	1
15	Identify the following parts A, B, C, D from the given picture  a) A- Stigma, B- Style, C-Ovary, D- Pollen tube b) A- Pollen grain , B- Stigma, C- Male germ cell , D- Pollen tube c) A- Pollen grain , B- Style, C- Male germ cell , D- Pollen tube d) A- Pollen grain , B- Style, C- Pollen tube , D- Stigma	1
16	A cross between a tall plant (TT) and dwarf plant (tt) resulted in progeny that were all tall plants as: a) Tallness is the dominant trait b) Shortness is the dominant trait c) Tallness is the recessive trait d) Height of plant is not governed by gene t or t	1
	Q.no 17 to 20 are Assertion and Reason based questions. These consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below: a) Both (A) and (R) are true, and (R) is the correct explanation of (A) b) Both (A) and (R) are true, and (R) is not the correct explanation of (A) c) (A) is true but (R) is false d) (A) is False but (R) is true	
17	Assertion (A): The extraction of metals from their sulphide ores cannot take place without roasting of the ore. Reason (R): Roasting converts sulphide ores directly into metals.	1
18	Assertion(A): Most of the plants close their stomata at night. Reason(R): Closing of stomata helps to conserve water as large amount of water evaporates from the leaves.	1
19	Assertion(A): The rainbow is a natural spectrum of sunlight in the sky. Reason(R): Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.	1

20	Assertion(A): Higher the Refractive index of the medium, lesser will be the speed of light in that Medium. Reason(R): Refractive index is inversely proportional to the speed of light	1
	Section – B Question No. 21 to 26 are very short answer questions	
21.	a) What are the benefits of having a four-chambered heart in birds and mammals? b) Name the following parts in the given picture  OR a) Why is there a difference in the rate of breathing between aquatic organisms and terrestrial organisms? Explain. b) Name the respiratory pigment present in the Human blood.	2
22	a) State two functions of stomata. b) How do guard cells regulate the opening and closing of stomatal pore? 	2
23	Observe the given picture and answer the questions that follow:  a) Name the type of chemical reaction that takes place in the above given experiment and Name the products formed during the reaction. b) Write balanced chemical equation for the reaction.	2
24	A student has focused the image of an object of height 3cm on a white screen using a concave mirror of focal length 12cm. If the distance of the object from the mirror is 18cm, find the values of the following: a) Distance of the image from the mirror. b) Height of the image.	2
25	What is the difference between food chain and food web. Draw a simple aquatic food chain.	2

26	a) Why is the tungsten used exclusively as filament for almost all electric lamps? b) Why are the conductors of electric heating devices, such as bread-toasters and electric irons, made of an alloy rather than a pure metal? OR a) How does the resistance of a wire vary with its area of cross-section? b) Why Copper and Aluminium wires are usually used for electricity transmission?	2
Section – C Question No. 27 to 33 are short answer questions		
27	a) What is the first step in the breakdown of glucose during aerobic and anaerobic respiration? Where does it take place? b) ATP is called the energy currency of the cell. Why? c) What is meant by “residual volume of air” in a breathing cycle?	3
28	(a) Why did Mendel carry out an experiment to study inheritance of two traits in garden pea plant? (b) In a following cross between two heterozygous Round seeds $Rr \times Rr$, What will be the phenotypic ratio and genotypic ratio obtained in F1 generation?	3
29	Give reasons: a) Pickles are not stored in tin metal container. b) Ionic compounds are poor conductor of electricity in solid state. c) Carbonate and sulphide ores are usually converted into oxides during the process of extraction. OR a) What is thermite reaction? Write balanced chemical equation for the reaction involved.  b) Name the substance that are getting oxidized and reduced in the thermite reaction.	3
30	a) Identify the natural phenomenon shown in the picture given below.  b) Explain with the help of the above labelled diagram that why the position of the star is seen by us is not in its true position.	3
31	a) Define water of crystallization. b) Give the chemical formula for two compounds as example. c) How can it be proved that the water of crystallization makes a difference in the state and colour of the compounds?	1 1 1
32	a) State Ohm's Law. b) Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit? Justify your answer.	3
33	What is a solenoid? When does a solenoid behave as a magnet? Draw the pattern of the magnetic field produced inside it showing the directions of the magnetic field lines.	3

	Section – D	
	Question No. 34 to 36 are long answer questions	
34	a) An electrical device of resistance R is connected across a source of voltage V and draws a current I. Derive an expression for power in terms of current and resistance. b) Two electric bulbs rated 100 W, 220 V and 60 W, 220 V are connected in parallel to an electric mains of 220 V. Find the current draws by the bulbs from the mains. OR a) Define Power and state its SI unit. b) A torch bulb is rated 5V and 500 mA. Calculate its - Power - Resistance - Energy consumed when it is lighted for $2\frac{1}{2}$ hours	5
35	a) Name the following parts in human female reproductive system. - That produces egg - Where fusion of egg and sperm takes place - Where zygote is implanted b) Name and explain any two methods of contraception OR a) Why is fertilization not possible without pollination? b) State the post-fertilization changes that lead to fruit formation in plant. c) Write any two advantages of vegetative propagation.	5
36	a)i) Identify A, B, C and D $\text{A} \xrightarrow{\text{Alkaline KMnO}_4} \text{B}$ $\text{A} + \text{Na}_{(s)} \longrightarrow \text{CH}_3\text{CH}_2\text{ONa} + \text{H}_{2(g)}$ $\text{B} + \text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{Conc. H}_2\text{SO}_4} \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ $\text{B} + \text{NaOH} \longrightarrow \text{C}$ $\text{B} + \text{Baking soda} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{D (Colourless gas)}$ ii) What is Saponification? Write its Equation. iii) What change will Ram observe if he adds, soap solution to red and blue litmus paper? OR b) A compound 'X' on heating with excess conc. sulphuric acid at 443 K gives an unsaturated compound 'Y'. 'X' also reacts with Sodium metal to evolve a colourless gas 'Z'. Identify 'X', 'Y' and 'Z'. Explain the chemical equation involved in the formation of 'Y' and also Write the role of Sulphuric acid in the reaction.	2 2 1
	Section – E	
	Question No. 37 to 39 are Case Study based questions with 2 to 3 short sub-parts. Internal Choice is provided in one of these sub-parts.	
37	Read the following passage and answer the questions that follow. When various members of a class of organic compounds, like alkanes, alkenes and alkyne which can be represented by a general formula and have same functional group are arranged in order of increasing molecular weights, they are found to differ from each other by a fixed group of atoms. They are said to form a homologous series. All members of a particular homologous series can be represented by the same general formula. For example, alkanes can be represented by the general formulae $\text{C}_n\text{H}_{2n+2}$. Similarly, alkenes and alkynes series of homologues can be represented by the general formulae C_nH_{2n} and $\text{C}_n\text{H}_{2n-2}$ respectively.	2 + 2

	a) i) What is the difference in the molecular formula of any two consecutive members of a homologous series of organic compounds? ii) From the given below compounds find the compounds that belong to same homologous series? CH_4, C_4H_6, C_3H_6, C_5H_8 b) Write the next homologue of each of the following: i) C_2H_4 ii) C_4H_6 OR b) What are structural isomers? Give an example.													
38	Study the given figure of human brain and answer the questions that follow:  a) State the role of the spinal cord in reflex action. OR a) State the functions of cerebellum. b) What is the function of the fluid filled in the brain? c) Name the parts A and B.	2 + 1 + 1												
39	Read the following passage and answer the questions that follow. A student took three convex lenses of different focal lengths and performed the experiment to see the image formation by placing an object at different distances with these lenses as shown in the following table. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Case No.</th><th>Object distance</th><th>Focal length</th></tr> </thead> <tbody> <tr> <td>I</td><td>40 cm</td><td>20 cm</td></tr> <tr> <td>II</td><td>25 cm</td><td>15 cm</td></tr> <tr> <td>III</td><td>20 cm</td><td>30 cm</td></tr> </tbody> </table> a) List two properties of the image formed in Case I. b) In which one of the cases given in the table, the lens will form virtual image? c) Name the type of lens used by dentists. Give reason why do they use such type of lens. OR c) Find the power of the lens used in Case I.	Case No.	Object distance	Focal length	I	40 cm	20 cm	II	25 cm	15 cm	III	20 cm	30 cm	1 + 1 + 2
Case No.	Object distance	Focal length												
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